You said:

calculus hw 33

ChatGPT said:

Jerry is using **James Stewart, *Calculus*, 8th Edition**.

Do you mean **page 186–188, Problem 33**?

Please send a photo or the problem statement, and I'll walk through the solution step by step.

You said:

page 185 - 188, problem 30

Worked for 12s

ChatGPT said:

Calculus — Section 2.8, Problem 30 (Page 187)

James Stewart, Calculus, 8th Edition

Problem statement (paraphrased)

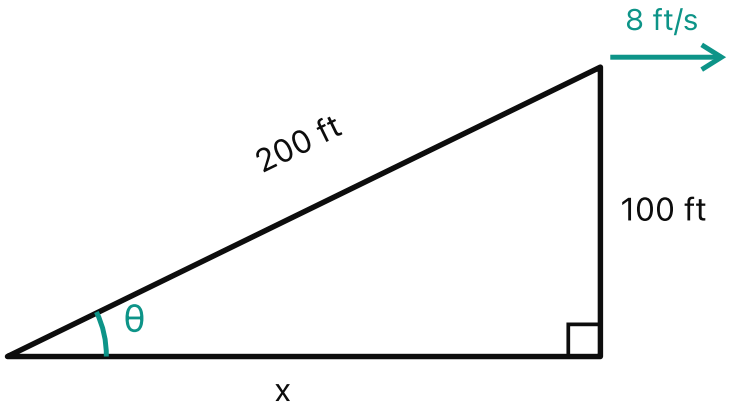
A kite flies horizontally at **8 ft/s**, maintaining an altitude of **100 ft**. Find how quickly the angle between the kite string and the horizontal is decreasing when the string is **200 ft long**. Studylib

Step 1: Define the variables

- x: horizontal distance from the person to the kite
- θ: angle between the string and the ground
- s: length of the kite string

We know:

$$\frac{dx}{dt} = 8 \text{ ft/s}, \quad s = 200 \text{ ft}$$



Step 2: Set up the equation

Using tangent:

$$\tan\theta = \frac{100}{x}$$

Differentiate both sides with respect to time:

$$\sec^2\theta \frac{d\theta}{dt} = -\frac{100}{x^2}\frac{dx}{dt}$$

Step 3: Find the horizontal distance

Using the Pythagorean theorem:

$$x^2 + 100^2 = 200^2$$

$$x^2 = 30000$$

Also,

$$\sec^2\theta = \frac{200^2}{x^2} = \frac{4}{3}$$

Step 4: Substitute

$$\frac{4}{3}\frac{d\theta}{dt} = -\frac{100}{30000}(8)$$

$$\frac{d\theta}{dt} = -0.02$$

Final answer

$$\boxed{\frac{d\theta}{dt} = -0.02 \text{ rad/s}}$$

The angle is **decreasing at 0.02 radians per second**.

The negative sign is important: the kite moves horizontally away from the person, so the angle between its string and the ground gets smaller.

You said:

problem 31 statement and solution

ChatGPT said:

Calculus — Section 2.8, Problem 31 (Page 187)

Textbook: James Stewart, *Calculus*, 8th Edition
Topic: Related Rates

Problem statement (paraphrased)

An equilateral triangle has sides that are increasing in length at a rate of **10 cm/min**. How fast is the area increasing when each side is **30 cm** long? GradeSaver

Step 1: Identify known values

Let:

- s = side length of the triangle (cm)

- A = area of the triangle (cm²)

Given:

$$\frac{ds}{dt} = 10 \text{ cm/min}$$

$$s = 30 \text{ cm}$$

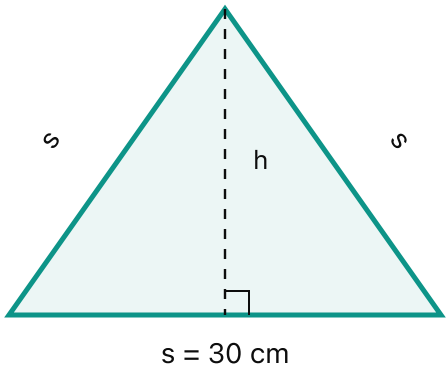
Find:

$$\boxed{\frac{dA}{dt} = ?}$$

Step 2: Write the area equation

For an equilateral triangle:

$$A = \frac{\sqrt{3}}{4}s^2$$



Step 3: Differentiate with respect to time

$$\frac{dA}{dt} = \frac{\sqrt{3}}{4}(2s)\frac{ds}{dt}$$

Simplify:

$$\boxed{\frac{dA}{dt} = \frac{\sqrt{3}}{2} s \frac{ds}{dt}}$$

Important: Since s changes with time, we must include $\frac{ds}{dt}$ by the chain rule.

Step 4: Substitute known values

$$\frac{dA}{dt} = \frac{\sqrt{3}}{2}(30)(10)$$

$$\frac{dA}{dt} = 150\sqrt{3}$$

Final answer

$$\boxed{\frac{dA}{dt} = 150\sqrt{3}\ \text{cm}^2/\text{min}}$$

Approximately:

$$\boxed{259.8\ \text{cm}^2/\text{min}}$$

Interpretation: When the triangle's sides are 30 cm long, its area is increasing at approximately 259.8 square centimeters per minute.

For Jerry: The key related-rates technique is to write the geometric formula first, differentiate with respect to time, and **only then substitute the given side length and rate**.

You said:

problem 33 statement and solution

ChatGPT said:

Calculus — Section 2.8, Problem 33 (Page 187)

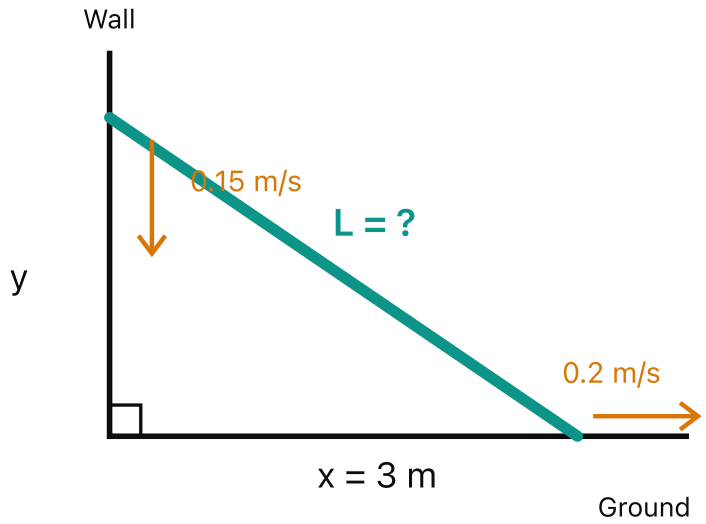
Textbook: James Stewart, *Calculus*, 8th Edition
Topic: Related Rates

Problem statement (paraphrased)

A ladder leans against a vertical wall. Its top slides downward at **0.15 m/s**. At the instant when the bottom is **3 m from the wall**, the bottom moves away at **0.2 m/s**.

Find the length of the ladder. GradeSaver +1

Step 1: Draw the diagram



Let:

- x = horizontal distance from the wall
- y = height of the top of the ladder
- L = length of the ladder (constant)

Given:

$$x = 3, \quad \frac{dx}{dt} = 0.2$$

$$\frac{dy}{dt} = -0.15$$

The negative sign means the top is moving downward.

Step 2: Write the Pythagorean equation

$$\boxed{x^2 + y^2 = L^2}$$

Differentiate both sides with respect to time:

$$2x\frac{dx}{dt} + 2y\frac{dy}{dt} = 2L\frac{dL}{dt}$$

Since the ladder's length does not change,

$$\frac{dL}{dt} = 0$$

Therefore,

$$\boxed{x\frac{dx}{dt} + y\frac{dy}{dt} = 0}$$

Step 3: Substitute the known values

$$3(0.2) + y(-0.15) = 0$$

$$0.6 - 0.15y = 0$$

$$0.15y = 0.6$$

$$\boxed{y = 4\ \text{m}}$$

Step 4: Find the ladder length

Return to the Pythagorean theorem:

$$L^2 = x^2 + y^2$$

$$L^2 = 3^2 + 4^2$$

$$L^2 = 9 + 16 = 25$$

$$\boxed{L = 5\ \text{m}}$$

Final answer

$$\boxed{5\ \text{meters}}$$

Key point for Jerry: Unlike most related-rates problems, Problem 33 asks for a *length*, not a rate. We use the two given rates to determine the missing height, then use the Pythagorean theorem to calculate the ladder's length.

Also, remember that the ladder's length is constant, so $\frac{dL}{dt} = 0$, even though both x and y are changing.

You said:

problem 41 statement and solution

ChatGPT said:

Calculus — Section 2.8, Problem 41 (Page 187)

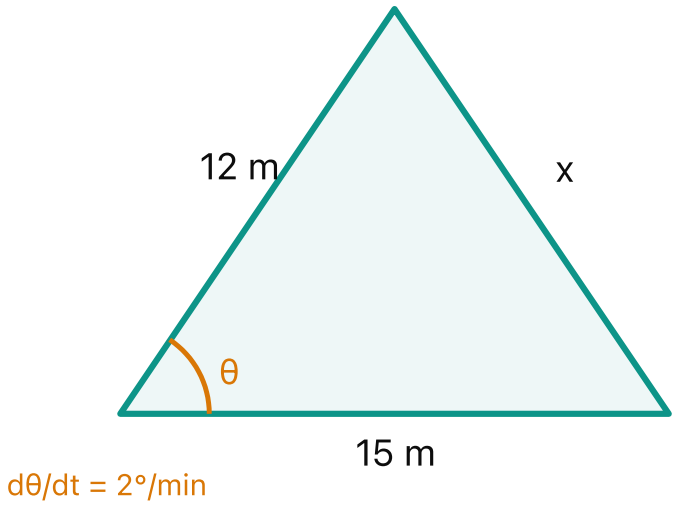
Textbook: James Stewart, *Calculus*, 8th Edition
Topic: Related Rates — Law of Cosines

Problem statement (paraphrased)

Two sides of a triangle have fixed lengths of **12 m** and **15 m**. The angle between these sides is increasing at a rate of **2° per minute**.

How fast is the length of the third side increasing when the angle between the fixed sides is **60°**? GradeSaver

Step 1: Draw and label the triangle



Let:

- θ = angle between the fixed sides
- x = length of the third side

Given:

$$\frac{d\theta}{dt} = 2^\circ/\text{min}$$

Convert degrees to radians:

$$\frac{d\theta}{dt} = 2 \cdot \frac{\pi}{180} = \boxed{\frac{\pi}{90}\ \text{rad/min}}$$

We want $\frac{dx}{dt}$ when $\theta = 60^\circ = \frac{\pi}{3}$.

Step 2: Apply the Law of Cosines

$$x^2 = a^2 + b^2 - 2ab\cos\theta$$

Substitute $a = 12$, $b = 15$:

$$x^2 = 12^2 + 15^2 - 2(12)(15)\cos\theta$$

$$\boxed{x^2 = 369 - 360\cos\theta}$$

Step 3: Differentiate with respect to time

$$2x\frac{dx}{dt} = 360\sin\theta\frac{d\theta}{dt}$$

Solve for $\frac{dx}{dt}$:

$$\boxed{\frac{dx}{dt} = \frac{180\sin\theta}{x}\frac{d\theta}{dt}}$$

Step 4: Find x when $\theta = 60^\circ$

$$x^2 = 369 - 360\cos 60^\circ$$

$$x^2 = 369 - 360\left(\frac{1}{2}\right)$$

$$x^2 = 189$$

$$\boxed{x = 3\sqrt{21}\ \text{m}}$$

Step 5: Substitute and calculate

$$\frac{dx}{dt} = \frac{180\sin 60^\circ}{3\sqrt{21}} \cdot \frac{\pi}{90}$$

Using $\sin 60^\circ = \frac{\sqrt{3}}{2}$:

$$\frac{dx}{dt} = \frac{180(\sqrt{3}/2)}{3\sqrt{21}} \cdot \frac{\pi}{90}$$

Simplify:

$$\frac{dx}{dt} = \frac{\pi\sqrt{3}}{3\sqrt{21}} = \frac{\pi}{3\sqrt{7}}$$

Final answer

$$\boxed{\frac{dx}{dt} = \frac{\pi}{3\sqrt{7}} \approx 0.396\ \text{m/min}}$$

The third side is **increasing at approximately 0.396 meters per minute**.

Key concepts for Jerry

1. **Law of Cosines:** Use it when two sides and their included angle are involved, rather than the Pythagorean theorem.
2. **Chain Rule:** Differentiating $\cos\theta$ with respect to time produces $-\sin\theta\frac{d\theta}{dt}$.
3. **Angle units:** Convert degrees per minute to radians per minute before using the derivative formula.
4. **Substitute after differentiating:** Keep x and θ as variables until you have the related-rates equation.

You said:

could you help me to derive Law of Cosines, sometimes I might forget it

ChatGPT said:

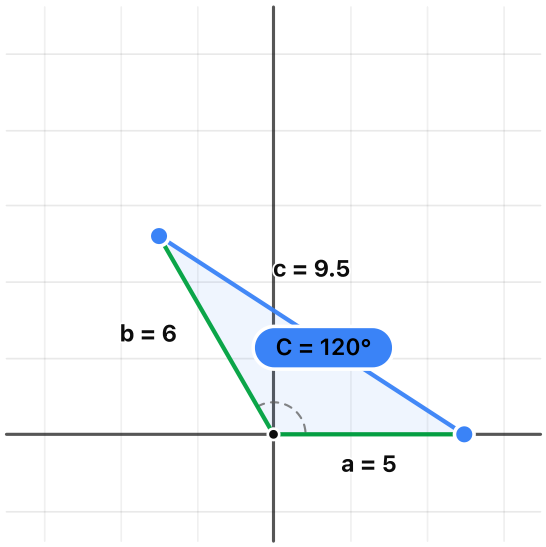
The **Law of Cosines** is actually just the **Pythagorean theorem plus a little trigonometry**. If you remember how to draw a perpendicular height, you can derive it anytime instead of memorizing the formula.

1. Start with any triangle

$$c^2 = a^2 + b^2 - 2ab\cos(C)$$

$$c = \sqrt{5^2 + 6^2 - 2(5)(6)\cos(120^\circ)} = 9.54$$

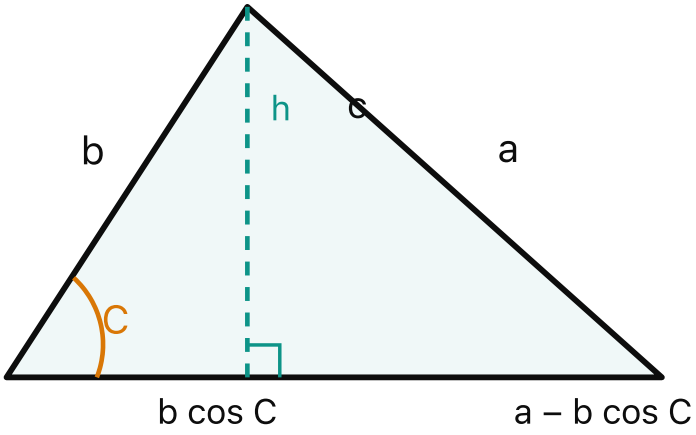
a	5
b	6
C	120°



Give feedback

We have two known sides a and b, with included angle C, and we want to find the opposite side c.

2. Draw a perpendicular height



The left small triangle gives us:

$$\cos C = \frac{\text{horizontal leg}}{b}$$

So the horizontal leg is:

$$b\cos C$$

And from sine:

$$h = b\sin C$$

The horizontal leg of the right triangle is therefore:

$$a - b\cos C$$

3. Apply the Pythagorean theorem

Look at the right triangle formed by the height, the remaining base, and side c.

$$c^2 = h^2 + (a - b\cos C)^2$$

Substitute $h = b\sin C$:

$$c^2 = (b\sin C)^2 + (a - b\cos C)^2$$

Expand:

$$c^2 = b^2\sin^2 C + a^2 - 2ab\cos C + b^2\cos^2 C$$

Rearrange:

$$c^2 = a^2 + b^2(\sin^2 C + \cos^2 C) - 2ab\cos C$$

Remember the identity:

$$\sin^2 C + \cos^2 C = 1$$

Therefore:

$$\boxed{c^2 = a^2 + b^2 - 2ab\cos C}$$

That's the Law of Cosines!

4. A useful way to remember it

Compare it with the Pythagorean theorem:

Pythagorean theorem	Law of Cosines
$c^2 = a^2 + b^2$	$c^2 = a^2 + b^2 - 2ab\cos C$

Only right triangles	Any triangle

When $C = 90^\circ$, we know $\cos 90^\circ = 0$, so:

$$c^2 = a^2 + b^2 - 2ab(0) = a^2 + b^2$$

The Law of Cosines becomes exactly the Pythagorean theorem.

5. Apply it to Jerry's Problem 41

For $a = 12$, $b = 15$, and $C = 60^\circ$:

$$c^2 = 12^2 + 15^2 - 2(12)(15)\cos 60^\circ$$

$$c^2 = 144 + 225 - 180 = 189$$

$$\boxed{c = 3\sqrt{21}}$$

My suggestion for Jerry: Rather than memorizing the entire derivation, remember these three things:

1. Draw a perpendicular height.
2. Express the height and horizontal projection using sine and cosine.
3. Apply the Pythagorean theorem and simplify using $\sin^2 C + \cos^2 C = 1$.

If you can do those three steps, you can reconstruct the Law of Cosines even during a calculus exam.

You said:

problem 43 statement and solution

ChatGPT said:

Calculus — Section 2.8, Problem 43 (Page 187)

Textbook: James Stewart, *Calculus*, 8th Edition
Topic: Related Rates — Rocket and Television Camera

Problem statement (paraphrased)

A television camera is positioned **4,000 ft** from the base of a rocket launching pad. The camera must adjust its angle of elevation to keep the rising rocket in view and its focus as the distance changes.

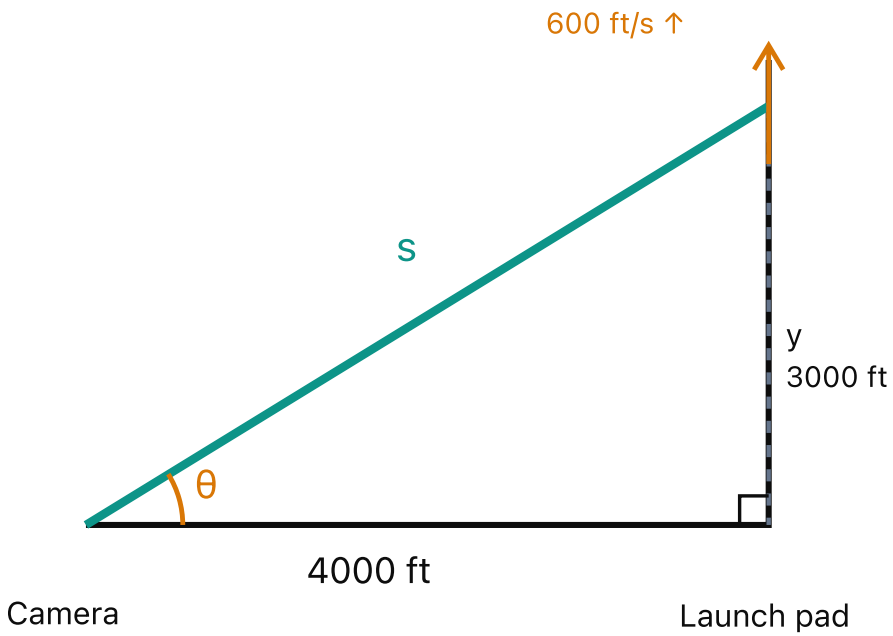
The rocket rises vertically and has a speed of **600 ft/s** when it reaches a height of **3,000 ft**.

Find:

- **(a)** How quickly is the distance between the camera and rocket increasing?
- **(b)** How quickly is the camera's angle of elevation increasing?

GradeSaver +1

Step 1: Draw the diagram



Let:

- y = rocket height (ft)

- s = distance from camera to rocket (ft)
- θ = camera's angle of elevation

Given:

$$y = 3000, \qquad \frac{dy}{dt} = 600 \text{ ft/s}$$

The horizontal distance of 4,000 ft is constant.

Part (a): Rate of change of distance

Step 2: Apply the Pythagorean theorem

$$s^2 = 4000^2 + y^2$$

Differentiate both sides with respect to time:

$$2s\frac{ds}{dt} = 2y\frac{dy}{dt}$$

Simplify:

$$\boxed{\frac{ds}{dt} = \frac{y}{s}\frac{dy}{dt}}$$

Step 3: Find s

When $y = 3000$:

$$s = \sqrt{4000^2 + 3000^2}$$

$$s = \sqrt{25,000,000} = 5000 \text{ ft}$$

Step 4: Substitute

$$\frac{ds}{dt} = \frac{3000}{5000}(600)$$

$$\boxed{\frac{ds}{dt} = 360 \text{ ft/s}}$$

Answer (a): The distance from the camera to the rocket is increasing at **360 ft/s**.

Part (b): Rate of change of angle

Step 5: Use tangent

From the triangle:

$$\tan\theta = \frac{y}{4000}$$

Differentiate with respect to time:

$$\sec^2\theta \frac{d\theta}{dt} = \frac{1}{4000}\frac{dy}{dt}$$

Solve:

$$\boxed{\frac{d\theta}{dt} = \frac{1}{4000\sec^2\theta}\frac{dy}{dt}}$$

Step 6: Find $\sec^2\theta$

From the triangle:

$$\cos\theta = \frac{4000}{5000} = \frac{4}{5}$$

Therefore:

$$\sec^2\theta = \frac{1}{\cos^2\theta} = \frac{25}{16}$$

Step 7: Substitute

$$\frac{d\theta}{dt} = \frac{600}{4000(25/16)}$$

$$= \frac{600(16)}{100000}$$

$$\boxed{\frac{d\theta}{dt} = 0.096 \text{ rad/s}}$$

Answer (b): The camera's angle of elevation is increasing at **0.096 radians per second**.

Final answers

(a) Distance rate	360 ft/s
(b) Angular rate	0.096 rad/s

Important observation for Jerry

Although the rocket moves vertically at 600 ft/s, the camera-to-rocket distance increases at only 360 ft/s at that instant.

This is because the camera measures the **diagonal distance**, not just the vertical height.

Also, notice that both quantities are increasing, so both derivatives are positive.

Calculus tip: In related-rates problems involving an angle, the most useful equations are often:

$$\tan\theta = \frac{\text{opposite}}{\text{adjacent}}$$

and

$$\sec^2\theta = 1 + \tan^2\theta$$

These let you find angular rates without explicitly calculating θ.

You said:

have to remember tan(x) derivative

ChatGPT said:

Yes! For Jerry's calculus class, **the derivative of $\tan x$ is one of the basic formulas he should memorize**.

$$\boxed{\frac{d}{dx}(\tan x) = \sec^2 x}$$

But it's also easy to derive if you forget it.

Derivation using the Quotient Rule

Start with:

$$\tan x = \frac{\sin x}{\cos x}$$

Apply the Quotient Rule:

$$\frac{d}{dx}\tan x = \frac{\cos x \cdot \cos x - \sin x(-\sin x)}{\cos^2 x}$$

$$= \frac{\cos^2 x + \sin^2 x}{\cos^2 x}$$

Since $\sin^2 x + \cos^2 x = 1$:

$$= \frac{1}{\cos^2 x} = \boxed{\sec^2 x}$$

Important for related-rates problems

When the angle θ changes with time, apply the Chain Rule:

$$\boxed{\frac{d}{dt}\tan\theta = \sec^2\theta \frac{d\theta}{dt}}$$

This is exactly what we used in Problem 43.

Basic trigonometric derivatives to memorize

Function	Derivative
$\sin x$	$\cos x$
$\cos x$	$-\sin x$
$\tan x$	$\sec^2 x$
$\cot x$	$-\csc^2 x$
$\sec x$	$\sec x \tan x$
$\csc x$	$-\csc x \cot x$

My recommendation: Jerry should memorize all six, particularly the first three. Understanding how to derive $\tan' x$ from sine and cosine is a good backup during a test.

One more important detail: These derivative formulas assume angles are measured in **radians**, not degrees.

You said:

tanx = sinx/cosx = (sin'x cosx -sinxcos'x)/cos^2x

ChatGPT said:

Exactly right! You've applied the **Quotient Rule** correctly.

Starting with:

$$\tan x = \frac{\sin x}{\cos x}$$

Differentiate:

$$\frac{d}{dx}(\tan x) = \frac{(\sin x)' \cos x - \sin x (\cos x)'}{\cos^2 x}$$

Since

$$(\sin x)' = \cos x, \qquad (\cos x)' = -\sin x$$

Substitute:

$$= \frac{\cos x \cos x - \sin x(-\sin x)}{\cos^2 x}$$

$$= \frac{\cos^2 x + \sin^2 x}{\cos^2 x}$$

Using the identity $\sin^2 x + \cos^2 x = 1$:

$$= \frac{1}{\cos^2 x}$$

Therefore:

$$\boxed{\frac{d}{dx}(\tan x) = \sec^2 x}$$

This is an excellent way to reconstruct the derivative if you forget it during an exam. All you need to remember is the Quotient Rule and the derivatives of sine and cosine.

You said:

problem 44 statement and solution

ChatGPT said:

Calculus — Section 2.8, Problem 44 (Page 188)

Textbook: James Stewart, *Calculus*, 8th Edition
Topic: Related Rates — Lighthouse

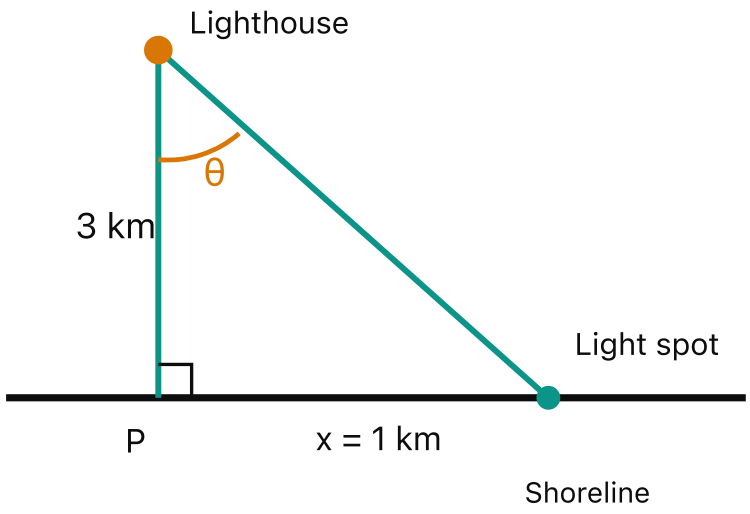
Problem statement (paraphrased)

A lighthouse is on a small island, **3 km from the nearest point P** on a straight shoreline.

Its light rotates **4 revolutions per minute**.

Question: How fast is the beam of light moving along the shoreline when it is **1 km from point P**? College Math T...

Step 1: Draw the diagram



Let:

- x = distance from point P to the light spot along the shoreline
- θ = angle between the beam and the perpendicular line from the lighthouse to P

Given:

$$x = 1 \text{ km}$$

$$\frac{d\theta}{dt} = 4 \text{ rev/min}$$

Find:

$$\boxed{\frac{dx}{dt} = ?}$$

Step 2: Convert revolutions to radians

One revolution equals 2π radians.

Therefore:

$$\frac{d\theta}{dt} = 4(2\pi)$$

$$\boxed{\frac{d\theta}{dt} = 8\pi \text{ rad/min}}$$

Step 3: Set up the trigonometric equation

From the right triangle:

$$\tan\theta = \frac{x}{3}$$

Or:

$$\boxed{x = 3\tan\theta}$$

Step 4: Differentiate with respect to time

Using the Chain Rule:

$$\frac{dx}{dt} = 3\sec^2\theta\frac{d\theta}{dt}$$

We know $\frac{d\theta}{dt} = 8\pi$, but we still need $\sec^2\theta$.

Step 5: Find $\sec^2\theta$

Remember the trigonometric identity:

$$\boxed{\sec^2\theta = 1 + \tan^2\theta}$$

Since $x = 1$ and the perpendicular distance is 3:

$$\tan\theta = \frac{1}{3}$$

Therefore:

$$\sec^2\theta = 1 + \left(\frac{1}{3}\right)^2$$

$$\boxed{\sec^2\theta = \frac{10}{9}}$$

Step 6: Substitute

$$\frac{dx}{dt} = 3\left(\frac{10}{9}\right)(8\pi)$$

Simplify:

$$\frac{dx}{dt} = \frac{80\pi}{3}$$

Final answer

$$\boxed{\frac{dx}{dt} = \frac{80\pi}{3}\ \text{km/min}}$$

$$\boxed{\frac{dx}{dt} \approx 83.78\ \text{km/min}}$$

The light spot moves along the shoreline at approximately **83.78 km/min**, or about **5,027 km/h**.

This high speed is possible because the light spot is not a physical object traveling along the shore.

Key concepts for Jerry

This problem is very similar to Problem 43 (the rocket and camera).

Problem 43	Problem 44
Rocket moves vertically	Light spot moves horizontally
Given dy/dt	Given $d\theta/dt$
Find $d\theta/dt$	Find dx/dt
$\tan\theta = y/4000$	$\tan\theta = x/3$

Both problems use the same derivative:

$$\boxed{\frac{d}{dt}(\tan\theta) = \sec^2\theta \frac{d\theta}{dt}}$$

Important: The two things to remember for Problem 44 are converting revolutions to radians and using $\sec^2\theta = 1 + \tan^2\theta$. You don't need to calculate the actual angle θ at all.

You said:

problem 45 statement and solution

ChatGPT said:

Calculus — Section 2.8, Problem 45 (Page 188)

Textbook: James Stewart, *Calculus*, 8th Edition
Topic: Related Rates — Airplane and Tracking Telescope

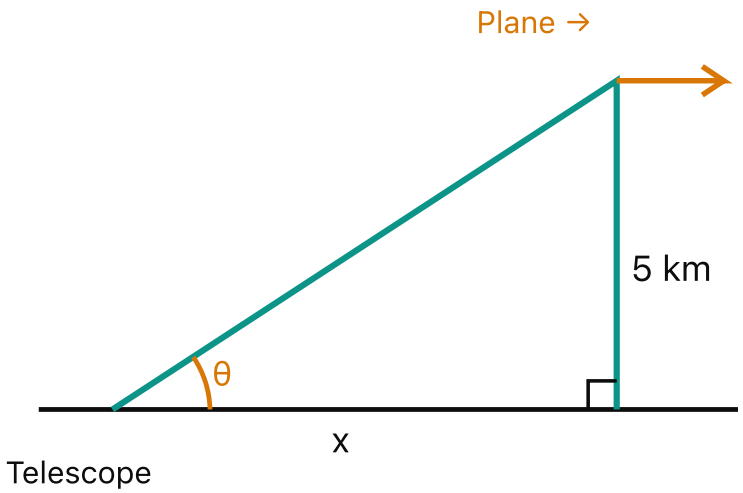
Problem statement (paraphrased)

An airplane flies horizontally at a constant altitude of **5 km**, passing directly over a tracking telescope on the ground.

When the telescope's angle of elevation is $\frac{\pi}{3}$ radians, the angle is decreasing at a rate of $\frac{\pi}{6}$ radians per minute.

Question: How fast is the airplane traveling at that instant? Exercise 45

Step 1: Draw the diagram



Let:

- x = horizontal distance between the airplane and telescope
- θ = angle of elevation

Given:

$$\theta = \frac{\pi}{3}$$

$$\frac{d\theta}{dt} = -\frac{\pi}{6}\ \text{rad/min}$$

The negative sign indicates that the angle is decreasing.

We want to find $\frac{dx}{dt}$.

Step 2: Set up the trigonometric relationship

From the right triangle:

$$\tan\theta = \frac{5}{x}$$

Solve for x:

$$x = \frac{5}{\tan\theta}$$

Since $\frac{1}{\tan\theta} = \cot\theta$:

$$\boxed{x = 5\cot\theta}$$

Step 3: Differentiate with respect to time

Remember:

$$\frac{d}{d\theta}\cot\theta = -\csc^2\theta$$

Apply the Chain Rule:

$$\boxed{\frac{dx}{dt} = -5\csc^2\theta\frac{d\theta}{dt}}$$

Step 4: Find $\csc^2\theta$

We know:

$$\theta = \frac{\pi}{3}$$

And:

$$\sin\frac{\pi}{3} = \frac{\sqrt{3}}{2}$$

Therefore:

$$\csc^2\frac{\pi}{3} = \frac{1}{\sin^2(\pi/3)} = \frac{1}{3/4} = \frac{4}{3}$$

Step 5: Substitute

$$\frac{dx}{dt} = -5\left(\frac{4}{3}\right)\left(-\frac{\pi}{6}\right)$$

$$= \frac{20\pi}{18}$$

$$\boxed{\frac{dx}{dt} = \frac{10\pi}{9}\ \text{km/min}}$$

Final answer

$$\boxed{\frac{10\pi}{9} \approx 3.49\ \text{km/min}}$$

The airplane is traveling at approximately **3.49 km/min**, equivalent to about **209.4 km/h**.

Key concept for Jerry

Problems 43, 44, and 45 all use trigonometry and the Chain Rule.

Problem	Trigonometric equation	Main derivative
43 — Rocket	$\tan\theta = y/4000$	$(\tan\theta)' = \sec^2\theta$
44 — Lighthouse	$x = 3\tan\theta$	$(\tan\theta)' = \sec^2\theta$
45 — Airplane	$x = 5\cot\theta$	$(\cot\theta)' = -\csc^2\theta$

An alternative for Problem 45: If Jerry forgets the derivative of cotangent, he can start with $\tan\theta = 5/x$ and differentiate implicitly, using only the derivative of tangent.

$$\sec^2\theta\frac{d\theta}{dt}=-\frac{5}{x^2}\frac{dx}{dt}$$

This leads to the same answer and avoids memorizing another derivative formula.

You said:

$\cot\theta=-\csc$

for me , remembering this formula is hard, especially after long time

ChatGPT said:

I agree. Remembering all six trigonometric derivative formulas after months or years can be difficult.

My suggestion is to memorize fewer formulas and learn how to derive the others. This is especially useful in calculus.

1. For cotangent, remember its relationship to tangent

$$\cot x=\frac{1}{\tan x}$$

Using the reciprocal rule:

$$\frac{d}{dx}\cot x=-\frac{1}{\tan^2 x}\cdot\sec^2 x$$

Now simplify:

$$=-\frac{1/\cos^2 x}{\sin^2 x/\cos^2 x}$$

$$= -\frac{1}{\sin^2 x}$$

Therefore:

$$\boxed{\frac{d}{dx}\cot x = -\csc^2 x}$$

You can reconstruct this whenever you need it.

2. Memorize just two basic derivatives

$$\boxed{(\sin x)' = \cos x}$$

$$\boxed{(\cos x)' = -\sin x}$$

With these two, plus the Product Rule, Quotient Rule, and Chain Rule, you can derive all six trigonometric derivatives.

Derivative	How to reconstruct
$(\tan x)'$	Differentiate $\sin x / \cos x$
$(\cot x)'$	Differentiate $\cos x / \sin x$
$(\sec x)'$	Differentiate $1 / \cos x$
$(\csc x)'$	Differentiate $1 / \sin x$

3. For Problem 45, you can avoid cotangent entirely

Rather than using $x = 5\cot\theta$, use:

$$\tan\theta = \frac{5}{x}$$

Differentiate:

$$\sec^2\theta\frac{d\theta}{dt} = -\frac{5}{x^2}\frac{dx}{dt}$$

Now you only need the tangent derivative, which you already know how to derive.

The long-term learning strategy: Memorize the smallest set of fundamental identities and rules, then use them to reconstruct more complicated formulas. This reduces the burden on memory and strengthens mathematical understanding.

For test speed, though, Jerry should practice the six derivatives until they become familiar; understanding their derivations provides a reliable backup.

If you'd like, I can turn these into a small set of flashcards focused on reconstructing formulas rather than simply memorizing them.

You said:

cotx = cosx/sinx, cot'x = (-sinx*sinx - cosx * cosx)/sin^2x = -1/sin^2x, i would rather derive on the spot

ChatGPT said:

Exactly right! Your derivation is **100% correct**, and I think deriving it on the spot is a very reasonable approach.

Start with:

$$\cot x = \frac{\cos x}{\sin x}$$

Using the Quotient Rule:

$$(\cot x)' = \frac{(-\sin x)\sin x - \cos x(\cos x)}{\sin^2 x}$$

$$= \frac{-\sin^2 x - \cos^2 x}{\sin^2 x}$$

Since $\sin^2 x + \cos^2 x = 1$:

$$= -\frac{1}{\sin^2 x}$$

$$\boxed{(\cot x)' = -\csc^2 x}$$

Why I like your approach

You only need to remember three fundamental things:

1. $(\sin x)' = \cos x$
2. $(\cos x)' = -\sin x$
3. The Quotient Rule

Everything else follows logically.

In fact, **tan and cot are almost identical derivations**:

	Tangent	Cotangent
Definition	$\sin x / \cos x$	$\cos x / \sin x$
Numerator after differentiation	$+\sin^2 x + \cos^2 x$	$-\sin^2 x - \cos^2 x$
Result	$1/\cos^2 x$	$-1/\sin^2 x$

The only practical trade-off is speed. Reconstructing the derivative might take 15–30 seconds instead of recalling it instantly.

For long-term mathematical understanding, I prefer your approach too. You aren't relying on remembering an isolated formula years later; you can recover it from principles you understand.

You said:

_____.
miles from a trackin
n changes at a rate

ChatGPT said:

Related Rates — Missile Problem

First, the blank at the top:

$$\boxed{\text{Radians}}$$

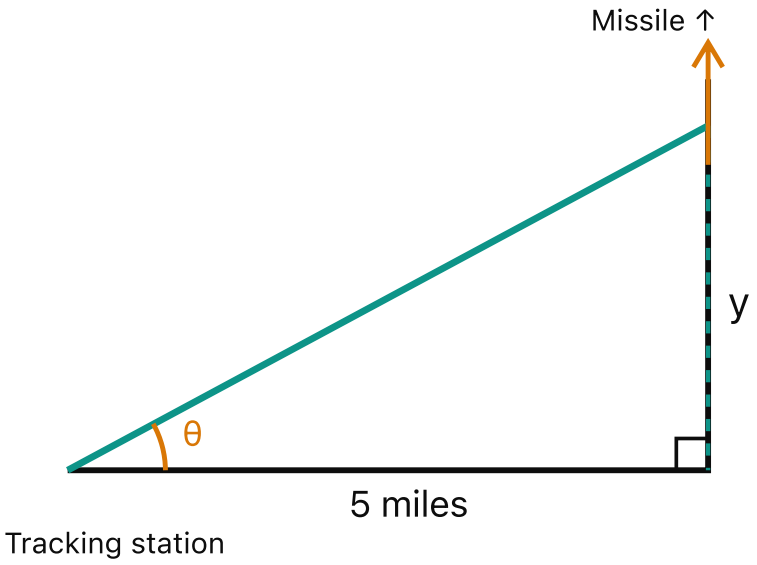
Trigonometric derivative formulas require angles to be measured in radians.

Problem 1: Given

A missile is launched vertically from a point 5 miles away from a tracking station.

- Horizontal distance: 5 miles (constant)
- Angle changes at $2^\circ/\text{s}$
- Find the missile's vertical velocity when $\theta = 30^\circ$.

Step 1: Draw the diagram



Step 2: Convert degrees to radians

Given:

$$\frac{d\theta}{dt} = 2^\circ/\text{s}$$

Since $180^\circ = \pi$ radians:

$$\boxed{\frac{d\theta}{dt} = \frac{\pi}{90}\ \text{rad/s}}$$

Step 3: Set up the equation

From the right triangle:

$$\tan\theta = \frac{y}{5}$$

Therefore:

$$\boxed{y = 5\tan\theta}$$

Step 4: Differentiate with respect to time

Using the Chain Rule:

$$\frac{dy}{dt} = 5\sec^2\theta\frac{d\theta}{dt}$$

Step 5: Substitute

When $\theta = 30^\circ = \pi/6$:

$$\sec^2 30^\circ = \frac{1}{\cos^2 30^\circ} = \frac{1}{3/4} = \frac{4}{3}$$

Therefore:

$$\frac{dy}{dt} = 5\left(\frac{4}{3}\right)\left(\frac{\pi}{90}\right)$$

$$= \frac{2\pi}{27}$$

Final answer

$$\boxed{\frac{dy}{dt} = \frac{2\pi}{27} \text{ miles/s}}$$

$$\boxed{\frac{dy}{dt} \approx 0.2327 \text{ miles/s}}$$

That's approximately **1,229 ft/s upward**.

One important observation

This problem is almost identical to Stewart's Problem 43 (rocket and camera), except the known rate is reversed:

	Stewart Problem 43	Missile problem
Given rate	Rocket's vertical speed	Angle's rate of change
Unknown	Angle's rate of change	Missile's vertical speed
Main equation	$\tan\theta = y/4000$	$\tan\theta = y/5$

For Jerry: The biggest potential mistake is substituting 2 directly for $d\theta/dt$. It must first be converted from degrees per second to radians per second.

You said:



ChatGPT said:

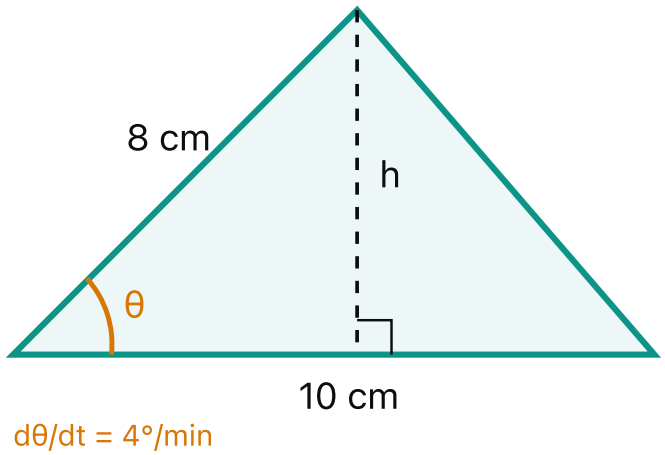
Problem 2 — Related Rates: Area of a Triangle

Problem statement

Two sides of a triangle are **8 cm and 10 cm** long. The angle between them is increasing at a rate of **4° per minute**.

Find how fast the area of the triangle is increasing when the angle between the two fixed sides is **45°**.

Step 1: Draw the triangle



Let A be the area and θ the angle between the fixed sides.

Given:

$$\frac{d\theta}{dt} = 4^\circ/\text{min}, \qquad \theta = 45^\circ$$

Find:

$$\frac{dA}{dt} = ?$$

Step 2: Derive the area formula

Instead of memorizing another formula, start with:

$$A = \frac{1}{2}(\text{base})(\text{height})$$

Choose the 10 cm side as the base.

Using sine, the height is:

$$h = 8\sin\theta$$

Therefore:

$$A = \frac{1}{2}(10)(8\sin\theta)$$

$$\boxed{A = 40\sin\theta}$$

Step 3: Differentiate with respect to time

Apply the Chain Rule:

$$\boxed{\frac{dA}{dt} = 40\cos\theta\frac{d\theta}{dt}}$$

Step 4: Convert degrees to radians

Because the derivative of sine assumes radians:

$$\frac{d\theta}{dt} = 4 \cdot \frac{\pi}{180}$$

$$\boxed{\frac{d\theta}{dt} = \frac{\pi}{45}\ \text{rad/min}}$$

Step 5: Substitute

When $\theta = 45^\circ$:

$$\cos 45^\circ = \frac{\sqrt{2}}{2}$$

So:

$$\frac{dA}{dt} = 40\left(\frac{\sqrt{2}}{2}\right)\left(\frac{\pi}{45}\right)$$

Simplify:

$$\frac{dA}{dt} = \frac{4\pi\sqrt{2}}{9}$$

Final answer

$$\boxed{\frac{dA}{dt} = \frac{4\pi\sqrt{2}}{9} \approx 1.974\ \text{cm}^2/\text{min}}$$

The triangle's area is **increasing at approximately 1.97 cm² per minute**.

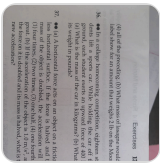
Key takeaway

This is closely related to Stewart's Problem 41, but there is an important difference:

- **Problem 41:** Find how fast the third side changes → use the Law of Cosines.
- **Problem 2:** Find how fast the area changes → use $A = \frac{1}{2}ab\sin\theta$.

Both formulas can be derived from basic geometry, so you don't need to rely on memorizing them.

You said:



solve problem 10 here then problem 11

ChatGPT said:

The photo shows **Problems 36 and 37 on page 133** of *Physics, 6th Edition* by Wilson, Buffa, and Lou, rather than Problems 10 and 11.

I'll solve the two visible problems in order.

Problem 36 — Mass and weight

Problem statement: Eighteen students lift a sports car. Each student exerts an upward force of 400 N.

(a) What is the mass of the car in kilograms?
(b) What is its weight in pounds?

Step 1: Find the total upward force

Since 18 students each exert 400 N:

$$F_{\text{total}} = 18(400) = 7200 \text{ N}$$

Assuming they are holding the car stationary, the upward force balances the car's weight:

$$mg = 7200$$

Step 2: Calculate mass

$$m = \frac{7200}{9.8}$$

$$\boxed{m \approx 735 \text{ kg}}$$

Step 3: Convert weight to pounds

Using $1 \text{ lb} \approx 4.448 \text{ N}$:

$$W = \frac{7200}{4.448}$$

$$\boxed{W \approx 1620 \text{ lb}}$$

Final answers: (a) approximately 735 kg; (b) approximately 1,620 lb.

If the teacher treats 400 N as one significant figure, the mass would instead be reported as approximately 700 kg. The intended precision depends on how the given value is interpreted.

Problem 37 — Newton's Second Law

Part (a)

Problem statement: A horizontal force acts on an object on a frictionless horizontal surface. If the force is halved and the object's mass is doubled, how does acceleration change?

Use Newton's Second Law:

$$F = ma$$

Rearrange:

$$\boxed{a = \frac{F}{m}}$$

The new force is $F/2$ and the new mass is $2m$.

Therefore:

$$a_{\text{new}} = \frac{F/2}{2m}$$

Simplify:

$$a_{\text{new}} = \frac{F}{4m}$$

But the original acceleration was:

$$a = \frac{F}{m}$$

Therefore:

$$\boxed{a_{\text{new}} = \frac{1}{4}a}$$

Answer: (4) one fourth as great.

Part (b)

Problem statement: If the acceleration of the object is 1.0 m/s², and the force on it is doubled and its mass is halved, what is the new acceleration?

Start with:

$$a = \frac{F}{m}$$

New force:

$$F_{\text{new}} = 2F$$

New mass:

$$m_{\text{new}} = \frac{m}{2}$$

Therefore:

$$a_{\text{new}} = \frac{2F}{m/2}$$

Dividing by a fraction means multiplying by its reciprocal:

$$a_{\text{new}} = 2F \cdot \frac{2}{m}$$

$$a_{\text{new}} = 4\frac{F}{m} = 4a$$

Given $a = 1.0\ \text{m/s}^2$:

$$\boxed{a_{\text{new}} = 4.0\ \text{m/s}^2}$$

Key concept for Jerry

Acceleration is **directly proportional to force** and **inversely proportional to mass**:

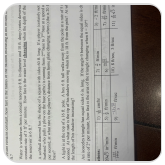
$$\boxed{a \propto \frac{F}{m}}$$

Change	Effect on acceleration
Double the force	Doubles
Halve the force	Halves
Double the mass	Halves
Halve the mass	Doubles
Halve force and double mass	Becomes $1/4$
Double force and halve mass	Becomes 4 times

This is the central concept behind Problem 37. Once Jerry understands $a = F/m$, he can answer these scaling questions without performing a full numerical calculation.

If you intended **Problems 10 and 11**, please send that page, since they aren't visible in this photo.

You said:



no, problem 10 in this picture:

ChatGPT said:

Problem 10 — Related Rates: Shadow of a Boy

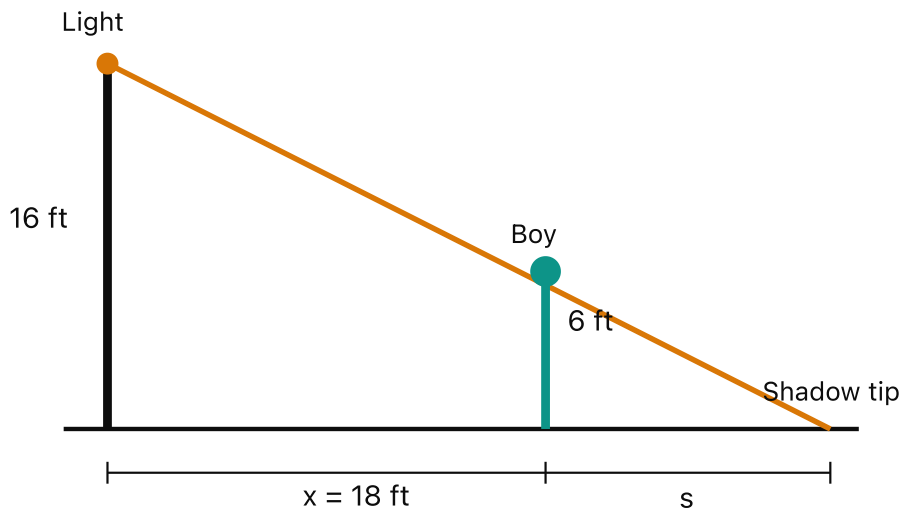
Problem statement (from your picture)

A light is at the top of a **16 ft pole**. A boy **6 ft tall** walks away from the pole at a rate of **5 ft per second**.

(a) At what rate is the tip of his shadow moving when he is 18 ft from the pole?

(b) At what rate is the length of his shadow increasing?

Step 1: Draw the diagram



Let:

- x = distance from the pole to the boy
- s = length of the boy's shadow
- $x + s$ = distance from the pole to the tip of the shadow

Given:

$$\frac{dx}{dt} = 5 \text{ ft/s}$$

Step 2: Use similar triangles

The large triangle has height 16 ft and base $x + s$.

The smaller triangle has height 6 ft and base s.

Therefore:

$$\frac{16}{x+s} = \frac{6}{s}$$

Cross-multiply:

$$16s = 6(x+s)$$

$$16s = 6x + 6s$$

$$10s = 6x$$

$$\boxed{s = \frac{3}{5}x}$$

Step 3: Differentiate with respect to time

$$\frac{ds}{dt} = \frac{3}{5}\frac{dx}{dt}$$

Substitute $dx/dt = 5$:

$$\frac{ds}{dt} = \frac{3}{5}(5)$$

$$\boxed{\frac{ds}{dt} = 3 \text{ ft/s}}$$

This answers **part (b)**.

Step 4: Find the speed of the shadow tip — part (a)

The tip is located at distance:

$$D = x + s$$

Differentiate:

$$\frac{dD}{dt} = \frac{dx}{dt} + \frac{ds}{dt}$$

$$= 5 + 3$$

$$\boxed{\frac{dD}{dt} = 8 \text{ ft/s}}$$

Final answers

(a) Speed of shadow tip	**8 ft/s**
(b) Rate of shadow length increase	**3 ft/s**

Notice that the problem gives the boy's distance as 18 ft, but we never needed to substitute it. That's because the similar triangles establish a constant proportional relationship between x and s.

Key point for Jerry: Distinguish between the speed of the shadow's tip (8 ft/s) and the rate at which the shadow itself grows (3 ft/s). They are not the same.

Would you like me to continue with **Problem 11**, the isosceles triangle problem immediately below it?

You said:

solve problem 11

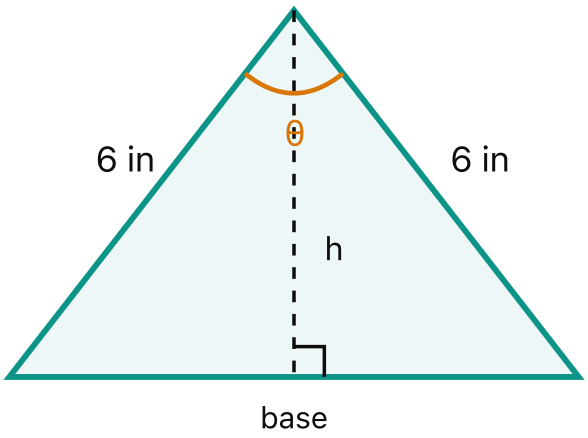
ChatGPT said:

Problem 11 — Related Rates: Area of an Isosceles Triangle

Problem statement

An isosceles triangle has equal sides **6 inches long**. If the angle θ between the equal sides is changing at a rate of **2° per minute**, how fast is the area of the triangle changing when $\theta = 30^\circ$?

Step 1: Draw the triangle



Given:

$$\frac{d\theta}{dt} = 2^\circ/\text{min}$$

Convert degrees to radians:

$$\frac{d\theta}{dt} = 2 \cdot \frac{\pi}{180} = \boxed{\frac{\pi}{90}\ \text{rad/min}}$$

We want $\frac{dA}{dt}$ when:

$$\theta = 30^\circ = \frac{\pi}{6}$$

Step 2: Derive the area formula

Instead of memorizing a special triangle area formula, begin with:

$$A = \frac{1}{2}(\text{base})(\text{height})$$

Choose one of the 6-inch sides as the base.

The perpendicular height relative to that side is:

$$h = 6\sin\theta$$

Therefore:

$$A = \frac{1}{2}(6)(6\sin\theta)$$

$$\boxed{A = 18\sin\theta}$$

This is an example of the general formula:

$$A = \frac{1}{2}ab\sin C$$

where C is the included angle between sides a and b.

Step 3: Differentiate with respect to time

Since $A = 18\sin\theta$:

$$\boxed{\frac{dA}{dt} = 18\cos\theta\frac{d\theta}{dt}}$$

Remember the Chain Rule: θ changes with time.

Step 4: Substitute

At $\theta = 30^\circ$:

$$\cos 30^\circ = \frac{\sqrt{3}}{2}$$

Therefore:

$$\frac{dA}{dt} = 18\left(\frac{\sqrt{3}}{2}\right)\left(\frac{\pi}{90}\right)$$

Simplify:

$$= 9\sqrt{3}\cdot\frac{\pi}{90}$$

$$= \frac{\pi\sqrt{3}}{10}$$

Final answer

$$\boxed{\frac{dA}{dt} = \frac{\pi\sqrt{3}}{10}\ \text{in}^2/\text{min}}$$

Approximately:

$$\boxed{0.544\ \text{in}^2/\text{min}}$$

The area is **increasing at approximately 0.544 square inches per minute**.

Important concept for Jerry

The key is recognizing that two fixed sides and their included angle determine the triangle's area:

$$\boxed{A = \frac{1}{2}ab\sin\theta}$$

This formula can always be reconstructed from $A = \frac{1}{2}bh$ using trigonometry.

Notice that as θ increases from 0° to 90°, the area increases, but at a decreasing rate. When $\theta = 90^\circ$, the area reaches its maximum and $dA/dt = 0$, assuming the angle continues increasing at the same rate.